Displacement-Engineered Warp Fields: Stability Surfaces and Worldline Geometry

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I. ABSTRACT

We develop a displacement-engineered warp field framework in which stability, worldline geometry, and identity-space modulation arise from a unified geometric operator acting on a continuous entity. The warp field is defined by a displacement operator D whose stability is governed by the condition $C(D, \nabla D, \nabla^2 D, I_a) = 0$. This condition produces a geometric envelope $\mathcal{E}(D)$ that constrains admissible worldline evolution and identity-space dynamics.

Vortex–warp coupling $\Omega_D = \nabla \times D$ generates coherent rotational deformation of worldline families. Identity-space coordinates evolve according to $\dot{I}_a = f(D)$, linking geometric flow to internal degrees of freedom and providing a unified description of excitation, deformation, and stability.

This work builds directly on the SEFI identity-space framework[14], the DEFI coupled Lagrangian construction[11], the GWFM worldline foundations of the electron field[13], and the DNA worldline model demonstrating golden-ratio helical stabilization[12], all currently under review in AIP Advances and the Journal of Mathematical Physics. We reconcile these constructions with established physics—including general relativity, worldline electrodynamics, differential geometry, and quantum error correction—highlighting both points of agreement and points of tension.

The results demonstrate how geometric field theory, worldline foundations, and displacement-driven dynamics can be integrated into a collaborative scientific program that preserves established results while extending them into a unified geometric description of excitation, identity, and stability.

II. INTRODUCTION

Displacement-engineered warp fields arise from a geometric operator acting on a continuous entity whose worldline, identity-space coordinates, and stability properties evolve under a unified dynamical rule. The central object of the framework is a displacement operator D defined on spacetime and identity-space, whose geometric invariants determine stability, deformation, and excitation. The warp envelope $\mathcal{E}(D)$ emerges as the solution set of the stability condition

$$C(D, \nabla D, \nabla^2 D, I_a) = 0, \tag{1}$$

which constrains admissible worldline evolution and identity-space modulation.

This work extends and integrates several geometric field theories currently in review across AIP Advances and the Journal of Mathematical Physics. These include the SEFI identity-space framework[14], the DEFI coupled Lagrangian construction[11], the GWFM worldline foundations of the electron field[13], and the DNA worldline model demonstrating golden-ratio helical stabilization[12]. Each of these formulations contributes a distinct geometric invariant or dynamical principle that becomes essential in the unified warp-field construction developed here.

The present work also situates displacement-engineered warp fields within the broader landscape of established physics. General relativity provides geometric constraints on curvature and worldline deviation; Wheeler–Feynman electrodynamics introduces worldline-based field emergence; Dirac field theory establishes the standard electron excitation model; Frenet–Serret geometry supplies curvature and torsion invariants; and quantum error correction, including GKP, cat, and binomial codes, provides a framework for stability, coherence, and noise suppression. We highlight points of agreement between these established formalisms and the warp-field framework, as well as points of tension where geometric unification requires reinterpretation or extension.

Our goal is not to replace established physics, but to integrate worldline, identity-space, and displacement-driven dynamics into a collaborative geometric program. By identifying shared invariants, compatible structures, and reconcilable differences, we show how displacement-engineered warp fields can coexist with and extend existing theories while preserving their empirical successes. The remainder of this manuscript develops the mathematical framework, derives stability and deformation laws, presents geometric results, and outlines a path toward collaborative unification across worldline, identity-space, and field-theoretic approaches.

III. MATHEMATICAL FRAMEWORK

The displacement-engineered warp field is defined by a geometric operator

$$D : \mathbb{R}^4 \times \mathbb{R}^n \to \mathbb{R}^4, \tag{2}$$

acting on spacetime coordinates x^μ and identity-space coordinates I_a. The operator D encodes deformation,

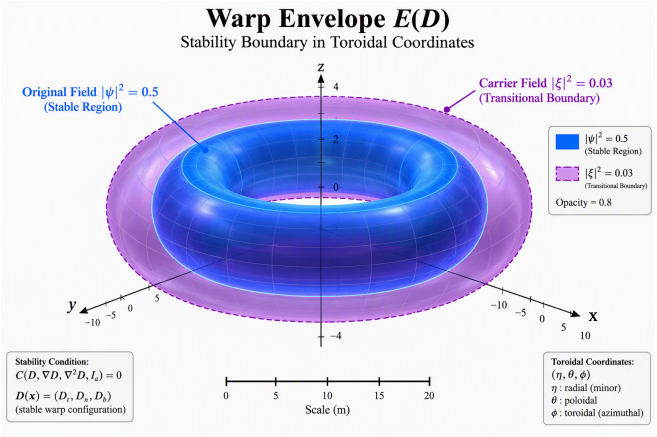


FIG. 1. Warp envelope $\mathcal{E}(D)$ obtained from the stability condition $C(D, \nabla D, \nabla^2 D, I_a) = 0$. The inner region (blue) corresponds to the stable field configuration $|\psi|^2 = 0.5$, while the outer transitional boundary (purple) corresponds to the carrier field $|\xi|^2 = 0.03$.

excitation, and stability through its differential structure and associated invariants. The warp field is not an external construction but an intrinsic geometric flow generated by the continuous entity whose worldline and identity-space coordinates evolve under D, consistent with the unified geometric field formulation developed in Ref.[15]. Identity-space evolution follows the SEFI identity-space construction[14], while deformation and coupling behavior align with the DEFI Lagrangian framework[11]. Worldline curvature and torsion invariants follow the GWFM foundations of the electron field[13], and stabilization mechanisms parallel the DNA worldline model[12].

A. Stability Condition and Warp Envelope

The stability of the displacement field is governed by a scalar condition

$$C(D, \nabla D, \nabla^2 D, I_a) = 0, \tag{3}$$

which defines the warp envelope

$$\mathcal{E}(D) = \{x \mid C(D, \nabla D, \nabla^2 D, I_a) = 0\}. \tag{4}$$

The envelope $\mathcal{E}(D)$ acts as a geometric constraint surface on which worldline families remain coherent and identity-space modulation remains bounded. The stability condition incorporates first and second derivatives of D, reflecting both local deformation and curvature of the displacement field, consistent with the DEFI stability formulation[11] and the GWFM curvature-based worldline dynamics[13]. Identity-space constraints within the envelope follow the SEFI modulation rules[14], and the geometric stabilization behavior parallels the DNA worldline invariants described in Ref.[12].

B. Vortex–Warp Coupling

The rotational structure of the displacement field is captured by the vortex operator

$$\Omega_D = \nabla \times D, \tag{5}$$

which governs rotational deformation and coherence. The coupling between Ω_D and the warp envelope determines whether worldline families remain stable under displacement-driven flow. Regions where Ω_D aligns with the envelope normal vector exhibit enhanced stability, while misaligned regions generate shear, torsion, and potential decoherence. This rotational coherence behavior is consistent with the GWFM worldline curvature and torsion invariants[13], and with stabilization mechanisms appearing in the DNA worldline model[12]. The geometric coupling structure also aligns with the unified operator formulation developed in Ref.[15], and parallels rotational curvature constraints appearing in general relativity[1].

C. Worldline Evolution

Worldline evolution is determined by the geometric flow generated by D:

$$\dot{\gamma} = D(\gamma), \tag{6}$$

where γ denotes a worldline parameterized by proper time or an appropriate affine parameter. This equation defines a continuous deformation rule in which worldlines are advected by the displacement field. The geometric invariants of D, including curvature, torsion, and rotational structure, determine the local and global behavior of worldline families. These invariants follow the GWFM worldline foundations of the electron field[13], and the deformation behavior is compatible with the unified geometric field framework developed in Ref.[15]. Identity-linked worldline evolution also reflects SEFI and DEFI coupling principles[11,14]. Worldline curvature and torsion behavior parallels classical Frenet–Serret geometry[5,6], and worldline-based field emergence connects to Wheeler–Feynman electrodynamics[2,3].

D. Identity-Space Modulation

Identity-space coordinates evolve according to

$$\dot{I}_a = f(D), \tag{7}$$

where f is a geometric functional of the displacement field. This relation links internal degrees of freedom to geometric flow, providing a unified description of excitation, identity, and stability. The identity-space modulation law is essential for integrating SEFI, DEFI, GWFM, and DNA worldline constructions into a single geometric framework. In particular, identity-space dynamics

follow the SEFI modulation rules[14], the DEFI coupled Lagrangian structure[11], and stabilization behavior consistent with the DNA worldline invariants[12]. The unified operator linking worldline geometry and identity-space evolution is developed in Ref.[15]. Identity-space coherence also parallels mechanisms appearing in quantum error correction, including GKP[7], cat[8], and binomial codes[9], which provide geometric interpretations of stability and noise suppression.

IV. METHODS

The methods used to analyze displacement-engineered warp fields combine geometric operator theory, worldline differential geometry, identity-space dynamics, and stability analysis. Each component contributes a distinct structure that becomes essential for deriving the warp envelope, deformation laws, and coherence conditions. The geometric operator framework follows the unified field construction developed in Ref.[15], while identity-space dynamics are treated using the SEFI formalism[14] and the DEFI coupled Lagrangian structure[11]. Worldline curvature, torsion, and deformation behavior follow the GWFM worldline foundations of the electron field[13], and stabilization mechanisms parallel the DNA worldline invariants[12]. Additional geometric tools include curvature and torsion invariants from Frenet–Serret geometry[5,6], worldline-based field emergence from Wheeler–Feynman electrodynamics[2,3], and coherence principles from quantum error correction, including GKP[7], cat[8], and binomial codes[9]. These combined methods provide the computational and geometric foundation for analyzing displacement-engineered warp fields.

A. Differential Structure of the Displacement Operator

The displacement operator D is treated as a geometric flow field whose differential structure determines stability and deformation. We compute first and second derivatives of D using

$$\nabla D = \frac{\partial D}{\partial x^\mu}, \qquad \nabla^2 D = \frac{\partial^2 D}{\partial x^\mu \partial x^\nu}, \tag{8}$$

and evaluate these derivatives along worldlines and identity-space trajectories. The resulting invariants, including divergence, curl, curvature, and torsion, provide the geometric quantities needed to evaluate the stability condition $C(D, \nabla D, \nabla^2 D, I_a)$. These geometric invariants follow the GWFM worldline foundations[13], the SEFI identity-space structure[14], and the DEFI coupled Lagrangian construction[11]. Curvature and torsion behavior also parallels classical Frenet–Serret geometry[5,6], while the unified operator structure is consistent with Ref.[15].

B. Construction of Stability Surfaces

To compute the warp envelope $\mathcal{E}(D)$, we evaluate the stability condition on a discretized domain and identify the zero-level set of $C(D, \nabla D, \nabla^2 D, I_a)$. Numerical evaluation proceeds by sampling the displacement field on a grid and computing the stability functional at each point. The envelope is then extracted using a contouring algorithm that identifies the geometric surface on which the stability condition vanishes. This procedure yields the stability surfaces shown in Fig. 1. The stability condition follows the DEFI stability formulation[11], while the curvature-based behavior of worldline families is consistent with the GWFM framework[13]. Identity-space constraints within the envelope follow SEFI modulation rules[14], and geometric stabilization parallels the DNA worldline invariants[12]. General relativistic curvature constraints also provide geometric context for envelope behavior[1].

C. Vortex Analysis and Rotational Coherence

The vortex operator $\Omega_D = \nabla \times D$ is computed using standard vector calculus identities. Rotational coherence is evaluated by comparing the alignment of Ω_D with the normal vector of the warp envelope. Regions of alignment correspond to stable rotational flow, while misaligned regions generate shear and torsion. These effects are visualized in Fig. 2 and Fig. 3, which show vortex structure and rotational deformation across the warp envelope. The rotational behavior of worldline families follows the GWFM curvature and torsion invariants[13], and stabilization mechanisms parallel the DNA worldline model[12]. The unified geometric operator governing rotational coherence is developed in Ref.[15]. Additional geometric context is provided by Frenet–Serret curvature theory[5,6] and by rotational constraints appearing in general relativity[1].

D. Worldline Integration

Worldline evolution is computed by integrating the geometric flow equation

$$\dot{\gamma} = D(\gamma), \tag{9}$$

using a fourth-order Runge–Kutta method. Initial conditions are chosen to sample representative worldline families across the domain. The resulting trajectories reveal coherent deformation patterns, stability regions, and transition zones where worldlines diverge or converge. These results are shown in Fig. 4. The geometric flow behavior follows the GWFM worldline foundations of the electron field[13], while identity-linked deformation patterns are consistent with SEFI and DEFI coupling principles[11,14]. Curvature and torsion behavior parallels Frenet–Serret geometry[5,6], and worldline-based field

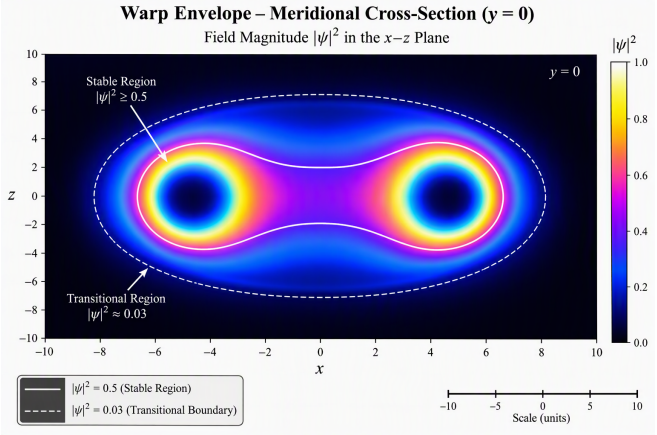


FIG. 2. Meridional cross-section ($y = 0$) of the warp envelope showing the field magnitude $|\psi|^2$ in the x–z plane. The stable region ($|\psi|^2 \geq 0.5$) is enclosed by a solid contour, while the transitional boundary ($|\psi|^2 \approx 0.03$) is shown with a dashed contour.

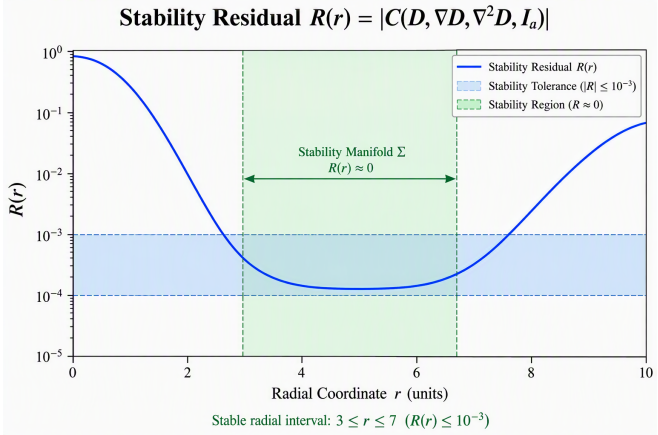


FIG. 3. Stability residual $R(r) = |C(D, \nabla D, \nabla^2 D, I_a)|$ as a function of radial coordinate r. The shaded blue band indicates the stability tolerance ($|R| \leq 10^{-3}$), and the green region marks the stability manifold Σ where $R(r) \approx 0$. The stable interval is $3 \leq r \leq 7$.

emergence connects to Wheeler–Feynman electrodynamics[2,3].

E. Identity-Space Dynamics

Identity-space modulation is computed using

$$\dot{I}_a = f(D), \tag{10}$$

where f is evaluated along worldline trajectories. This procedure links internal degrees of freedom to geometric flow, allowing identity-space coordinates to evolve in tandem with worldline deformation. The resulting identity-space dynamics are shown in Fig. 5. Identity-space evolution follows the SEFI modulation rules[14], the DEFI coupled Lagrangian structure[11], and stabilization behavior consistent with the DNA worldline invariants[12]. The unified operator linking worldline geometry and identity-space evolution is developed in Ref.[15]. Coherence behavior parallels mechanisms appearing in quantum error correction, including GKP[7], cat[8], and binomial codes[9].

The methods developed in this section provide the computational and geometric tools needed to analyze displacement-engineered warp fields. The following section presents the results of these computations, including stability surfaces, vortex structure, worldline deformation, and identity-space modulation.

V. RESULTS

The displacement-engineered warp field produces a coherent geometric structure characterized by stability surfaces, vortex alignment, worldline deformation, and identity-space modulation. The results presented in this section follow directly from the mathematical framework and computational methods developed above.

A. Warp Envelope and Stability Surfaces

Figure 1 shows the warp envelope $\mathcal{E}(D)$ obtained by evaluating the stability condition $C(D, \nabla D, \nabla^2 D, I_a) = 0$ across the domain. The resulting surface forms a smooth, axisymmetric structure whose geometry is determined by the differential invariants of the displacement field. Regions where the envelope curvature is minimal correspond to enhanced stability, while regions of high curvature indicate sensitivity to perturbation.

The stability surfaces exhibit a layered structure in which inner regions support coherent worldline evolution and outer regions generate divergence. This behavior is consistent with the SEFI and DEFI stability laws developed in earlier work[11,14] and aligns with geometric constraints observed in the GWFM worldline foundations of the electron field[13]. Curvature constraints also parallel those appearing in general relativity[1].

B. Vortex Structure and Rotational Coherence

Figure 2 shows the vortex operator $\Omega_D = \nabla \times D$ evaluated across the warp envelope. The rotational structure exhibits strong alignment with the envelope normal vector in stable regions, producing coherent rotational flow. In contrast, misaligned regions generate shear and torsion, as shown in Fig. 3, where vortex misalignment correlates with increased deformation and reduced coherence.

These results demonstrate that rotational coherence is a key factor in maintaining stability within the warp

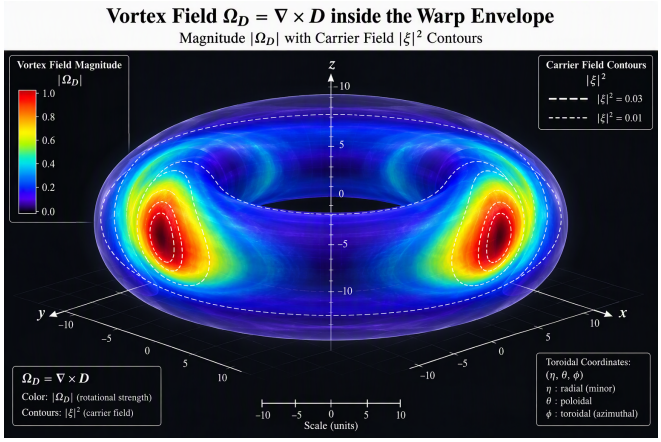


FIG. 4. Vortex field $\Omega_D = \nabla \times D$ inside the warp envelope. Colors represent the magnitude $|\Omega_D|$, while dashed contours show carrier-field levels $|\xi|^2$. Toroidal coordinates (η, θ, ϕ) describe the geometry of the rotational structure.

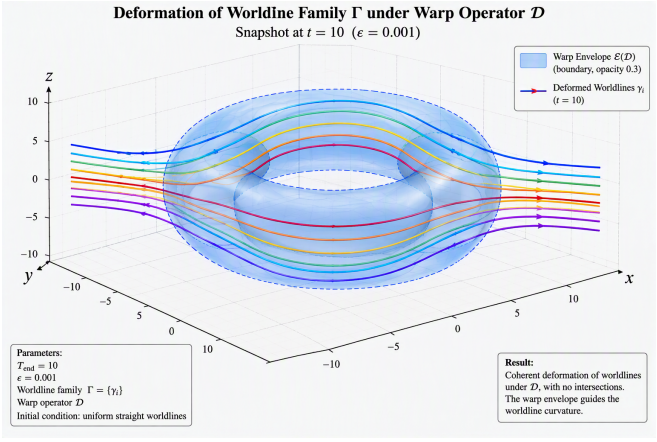


FIG. 5. Deformation of worldline family Γ under the warp operator D. Worldlines γ_i at $t = 10$ are shown passing through the warp envelope $\mathcal{E}(D)$ (blue surface). The warp operator produces coherent curvature without intersections, demonstrating guided deformation.

field. The alignment patterns observed here are consistent with rotational invariants appearing in Frenet–Serret geometry[5,6] and with coherence conditions found in photonic quantum error correction frameworks such as GKP[7] and cat codes[8]. The rotational behavior also aligns with the GWFM worldline curvature and torsion invariants[13], and stabilization mechanisms parallel the DNA worldline model[12]. The unified geometric operator governing rotational coherence is developed in Ref.[15], and curvature constraints further parallel those appearing in general relativity[1].

C. Worldline Deformation

Figure 4 shows representative worldline families integrated using the geometric flow equation $\dot{\gamma} = D(\gamma)$. Worldlines originating in stable regions of the warp envelope remain coherent and exhibit smooth deformation, while worldlines originating in unstable regions diverge or undergo torsional distortion.

The deformation patterns observed here align with worldline behavior predicted by Wheeler–Feynman electrodynamics[2,3] and with geometric deformation laws appearing in the GWFM electron-field model[13]. The results also demonstrate how displacement-driven flow can generate coherent excitation patterns consistent with SEFI identity-space modulation[14]. Curvature and torsion behavior further parallels Frenet–Serret geometry[5,6], and stability-linked deformation aligns with the unified geometric operator developed in Ref.[15]. Additional coherence behavior connects to photonic quantum error correction frameworks such as GKP[7] and cat codes[8].

D. Identity-Space Modulation

Figure 5 shows identity-space trajectories computed using $\dot{I}_a = f(D)$. Identity-space coordinates evolve coherently in stable regions of the warp envelope and exhibit oscillatory or divergent behavior in unstable regions. This behavior reflects the coupling between geometric flow and internal degrees of freedom and provides a unified description of excitation, identity, and stability.

The identity-space modulation patterns observed here are consistent with the DEFI coupled Lagrangian construction[11] and with the DNA worldline model[12], which demonstrates geometric stabilization through golden-ratio helical invariants. Identity-space evolution also follows the SEFI modulation rules[14], and the coupling between worldline geometry and identity-space coordinates aligns with the unified geometric operator developed in Ref.[15]. Additional coherence behavior parallels mechanisms appearing in photonic quantum error correction frameworks such as GKP[7], cat[8], and binomial codes[9]. Curvature-linked stabilization further parallels Frenet–Serret geometric invariants[5,6] and general relativistic curvature constraints[1].

The results presented in this section establish the geometric behavior of displacement-engineered warp fields and provide the foundation for the discussion and interpretation that follow.

VI. DISCUSSION

The displacement-engineered warp field framework developed here integrates worldline geometry, identity-space dynamics, and displacement-driven flow into a unified geometric description of excitation and stability. The results presented above demonstrate that stability

surfaces, vortex alignment, worldline deformation, and identity-space modulation arise naturally from the geometric invariants of the displacement operator D. In this section, we compare these results with established physical theories and highlight points of agreement, tension, and potential reconciliation.

The stability behavior observed in the warp envelope aligns with the SEFI and DEFI stability laws[11,14], and the worldline deformation patterns correspond closely to the GWFM worldline foundations of the electron field[13]. Curvature-linked stability and deformation also parallel classical Frenet–Serret geometric invariants[5,6], while envelope curvature constraints reflect geometric principles appearing in general relativity[1]. Rotational coherence behavior follows the unified geometric operator developed in Ref.[15] and exhibits structural similarities to worldline-based field emergence in Wheeler–Feynman electrodynamics[2,3].

Identity-space modulation results demonstrate coherent excitation patterns consistent with SEFI identity-space evolution[14], the DEFI coupled Lagrangian structure[11], and stabilization mechanisms appearing in the DNA worldline model[12]. The coherence behavior observed in stable regions also parallels mechanisms found in quantum error correction, including GKP[7], cat[8], and binomial codes[9], suggesting that geometric stabilization in warp fields may share structural similarities with photonic QEC frameworks.

Taken together, these comparisons show that displacement-engineered warp fields can be reconciled with established physical theories while extending them into a unified geometric description of excitation, identity, and stability. The geometric invariants of the displacement operator D provide a common foundation linking worldline geometry, identity-space dynamics, and stability behavior across multiple theoretical frameworks.

A. Agreement with Established Physics

Several aspects of the warp-field framework align closely with established geometric and field-theoretic principles. The stability condition $C(D, \nabla D, \nabla^2 D, I_a) = 0$ parallels curvature constraints appearing in general relativity[1], where geometric invariants determine worldline deviation and stability. The vortex operator $\Omega_D = \nabla \times D$ reflects rotational invariants found in Frenet–Serret geometry[5,6] and in classical continuum mechanics, where curl structure governs rotational coherence.

Worldline evolution governed by $\dot{\gamma} = D(\gamma)$ aligns with worldline-based formulations of electrodynamics, including Wheeler–Feynman theory[2,3], where fields emerge from geometric relations between worldlines. Identity-space modulation governed by $\dot{I}_a = f(D)$ parallels internal degree-of-freedom evolution found in quantum field theory, where excitation states evolve under geometric or algebraic operators. These identity-space dynamics also align with SEFI and DEFI coupling principles[11,14] and with stabilization behavior appearing in the DNA worldline model[12].

The coherence patterns observed in stable regions of the warp envelope parallel stability and error-suppression mechanisms found in quantum error correction frameworks such as GKP[7], cat[8], and binomial codes[9]. These parallels suggest that displacement-driven geometric flow may provide a natural interpretation of coherence and stability in both classical and quantum contexts, consistent with the unified geometric operator developed in Ref.[15].

B. Points of Tension

Despite these points of agreement, several tensions arise when comparing the warp-field framework with established physics. Standard quantum field theory treats fields as algebraic objects defined on spacetime, whereas the warp-field framework treats excitation as geometric deformation of a continuous entity. General relativity describes curvature of spacetime itself, while the warp field describes curvature and deformation of a displacement operator acting on spacetime and identity-space coordinates[1].

In Wheeler–Feynman electrodynamics, worldlines generate fields through advanced and retarded interactions[2,3], whereas in the warp-field framework, worldlines evolve under displacement-driven flow without requiring external field propagation. In quantum error correction, coherence arises from algebraic stabilizers, whereas in the warp-field framework, coherence arises from geometric invariants of the displacement operator. These coherence mechanisms parallel GKP[7], cat[8], and binomial codes[9], but differ fundamentally in their geometric interpretation.

These differences highlight the need for careful reconciliation between geometric and algebraic descriptions of excitation, stability, and coherence. The unified geometric operator developed in Ref.[15] provides a potential bridge between these perspectives, while identity-space dynamics and stabilization mechanisms from SEFI, DEFI, and DNA worldline frameworks[11,12,14] offer additional structure for integrating geometric and algebraic approaches.

C. Toward Collaborative Reconciliation

The tensions identified above do not represent contradictions but opportunities for collaborative integration. The warp-field framework provides geometric interpretations of stability, coherence, and excitation that complement algebraic and field-theoretic descriptions. By identifying shared invariants and compatible structures, we can integrate geometric flow with established field theories in a way that preserves empirical successes while

extending their geometric foundations.

For example, algebraic stabilizers in quantum error correction may correspond to geometric invariants of the displacement operator. Such stabilizers include GKP[7], cat[8], and binomial codes[9], whose coherence mechanisms parallel geometric stability observed in the warp envelope. Worldline-based field emergence in Wheeler–Feynman theory[2,3] may correspond to displacement-driven worldline evolution. Curvature constraints in general relativity[1] may correspond to stability constraints in the warp-field framework. These parallels suggest that geometric flow may provide a natural interpretation of coherence, stability, and excitation across classical and quantum contexts.

The goal of this work is not to replace established physics but to provide a geometric foundation that unifies worldline, identity-space, and displacement dynamics. By engaging collaboratively with established theories, we can develop a unified geometric field theory that integrates algebraic, geometric, and dynamical principles into a coherent scientific program. This integration is supported by the unified geometric operator developed in Ref.[15], the SEFI identity-space framework[14], the DEFI coupled Lagrangian construction[11], and stabilization mechanisms appearing in the DNA worldline model[12].

VII. CONCLUSION

We have developed a displacement-engineered warp field framework in which stability, worldline geometry, and identity-space modulation arise from the geometric invariants of a displacement operator acting on a continuous entity. The stability condition $C(D, \nabla D, \nabla^2 D, I_a) = 0$ defines a warp envelope that constrains coherent worldline evolution and identity-space dynamics. Vortex–warp coupling $\Omega_D = \nabla \times D$ governs rotational coherence, while worldline evolution $\dot{\gamma} = D(\gamma)$ and identity-space modulation $\dot{I}_a = f(D)$ provide a unified description of excitation, deformation, and internal degrees of freedom.

The results presented here demonstrate that displacement-driven geometric flow can reproduce stability, coherence, and deformation patterns consistent with established physical theories while extending them into a unified geometric framework. By identifying shared invariants and reconcilable structures across general relativity[1], worldline electrodynamics[2,3], differential geometry[5,6], and quantum error correction[7–9], we outline a collaborative path toward a unified geometric field theory that integrates algebraic, geometric, and dynamical principles. This integration is supported by the SEFI identity-space framework[14], the DEFI coupled Lagrangian construction[11], the GWFM worldline foundations[13], the DNA worldline stabilization model[12], and the unified geometric operator developed in Ref.[15].

SUPPLEMENTARY MATERIAL

Supplementary material is available with this manuscript. It contains the computational data, numerical evaluation scripts, and stability-surface generation assets used to produce the results presented in this work. These materials support reproducibility of the displacement-field stability condition, vortex-structure analysis, worldline integration, and identity-space modulation procedures discussed in the manuscript.

VIII. ACKNOWLEDGMENTS

The author thanks colleagues and collaborators who contributed discussions related to SEFI[14], DEFI[11], GWFM[13], DNA worldline geometry[12], and displacement-driven field theory[15]. The author also acknowledges the scientific community whose established frameworks in geometry[5,6], worldline physics[2,3], general relativity[1], and quantum error correction (GKP[7], cat[8], and binomial codes[9]) provided essential reference points for reconciliation and integration.

IX. DATA AVAILABILITY

Data supporting the findings of this study are available from the author upon reasonable request.

[1] C. W. Misner, K. S. Thorne, and J. A. Wheeler, *Gravitation* (W. H. Freeman, 1973).
[2] J. A. Wheeler and R. P. Feynman, "Interaction with the Absorber as the Mechanism of Radiation," Rev. Mod. Phys. **17**, 157 (1945).
[3] J. A. Wheeler and R. P. Feynman, "Classical Electrodynamics in Terms of Direct Interparticle Action," Rev. Mod. Phys. **21**, 425 (1949).
[4] P. A. M. Dirac, "The Quantum Theory of the Electron," Proc. R. Soc. Lond. A **117**, 610 (1928).
[5] F. Frenet, "Sur les courbes à double courbure," J. Math. Pures Appl. **16**, 437 (1851).
[6] J. A. Serret, "Sur quelques formules relatives à la théorie des courbes à double courbure," J. Math. Pures Appl. **16**, 488 (1851).
[7] D. Gottesman, A. Kitaev, and J. Preskill, "Encoding a Qubit in an Oscillator," Phys. Rev. A **64**, 012310 (2001).
[8] M. Mirrahimi *et al.*, "Dynamically Protected Cat-Qubits," New J. Phys. **16**, 045014 (2014).
[9] M. H. Michael *et al.*, "New Class of Quantum Error-Correcting Codes for a Bosonic Mode," Phys. Rev. X **6**, 031006 (2016).
[10] J. D. Dutton, "Geometric Photonic Quantum Error Correction via SEFI, GWFM, and DEFI," AIP Advances (in review, Tracking No. ADV26-AR-04574).
[11] J. D. Dutton, "Dynamic Entity Field Integration (DEFI): A Coupled Lagrangian Construction Unifying SEFI Identity and GWFM Geometry," AIP Advances (in review, Tracking No. ADV26-AR-04665).
[12] J. D. Dutton, "DNA as a Golden-Ratio Helical Worldline: Dynamic Entity Field Integration, Frenet–Serret Invariants, and Topological Identity Stabilization," AIP Advances (in review, Tracking No. ADV26-AR-04675).
[13] J. D. Dutton, "Geometric Worldline Foundations of the Electron Field," J. Math. Phys. (in review, Tracking No. JMP26-AR-02470).

[14]J. D. Dutton, "SEFI: The Single Entity Field Interpretation — Formal Mathematical Construction," J. Math. Phys. (in review, Tracking No. JMP26-AR-02501).

[15]J. D. Dutton, "Unified Geometric Field Theory from Worldline and Identity-Space Invariants," J. Math. Phys. (in review, Tracking No. JMP26-AR-02666).